

Geometric G: From Earth's Geometry to the Gravitational Constant

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Abstract

We present an empirical geometric observation: when the gravitational constant G is expressed in units derived directly from the Earth (meter from the meridian quadrant, kilogram from one liter of water, second from the mean solar day), the length and mass dimensions cancel with a factor of 10^3 (to 0.003%, limited by the density of water 999.97 kg/m^3), leaving a pure inverse square of time:

$$G_{\text{Earth}} = \frac{31^2 \times \sqrt{5}}{\phi^{12}} \times 10^{-8} \text{ s}^{-2} \approx 6.67354674496 \dots \times 10^{-8} \text{ s}^{-2}$$

In SI units this corresponds to a value lying between the two most precise laboratory measurements:

Source	Value ($\times 10^{-11}$)	Gap from G_{Earth}
Schlamminger (2014)	6.6729 ± 0.0005	+0.010%
G_{Earth} (this work)	$6.67354674496 \dots$	— — —
CODATA 2018	6.67430 ± 0.00015	−0.011%

The mutual disagreement between laboratories (0.021%) is larger than the deviation of our geometric value from either measurement. No CODATA values were used in the derivation.

1 Earth-Based Unit Definitions

Let M denote the length of one quarter of the Earth's meridian.

- **Meter:** $1 \text{ m} = \frac{M}{10^7}$
- **Decimeter:** $1 \text{ dm} = \frac{M}{10^8}$
- **Kilogram:** mass of 1 dm^3 of water
- **Second:** $1 \text{ s} = \frac{1}{86\,400}$ of the mean solar day

2 Cancellation of Length and Mass

From these definitions:

$$1 \text{ m}^3 = \frac{M^3}{10^{21}}, \quad 1 \text{ kg} \approx \frac{M^3}{10^{24}} \quad (\text{water density } 999.97 \text{ kg/m}^3)$$

$$\frac{1 \text{ m}^3}{1 \text{ kg}} = 10^3 \quad (\text{to } 0.003\%)$$

Applying this to the Kochenov geometric formula:

$$G_{\text{Koch}} = \frac{31^2 \sqrt{5}}{\phi^{12}} \times 10^{-11} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2}$$

Substituting $\text{m}^3/\text{kg} = 10^3$:

$$G_{\text{Earth}} = \frac{31^2 \sqrt{5}}{\phi^{12}} \times 10^{-8} \text{ s}^{-2}$$

All length and mass dependence cancels. Only the second remains.

3 A Geometric Observation: 720 and the Second

The number of seconds per mean solar day factorises as:

$$86\,400 = 120 \times 720 = 120 \times 6!$$

Independently, the golden ratio satisfies:

$$\frac{\phi^{12}}{\sqrt{5}} = 144 + \frac{1}{720} + \mathcal{O}(10^{-8})$$

The same integer 720 appears both in the Earth's daily rotation and in the geometric expansion of $\phi^{12}/\sqrt{5}$. This is noted as an empirical observation, not a proof.

4 What This Is Not

This is an empirical observation. No dynamical model is claimed. The factor $\text{m}^3/\text{kg} = 10^3$ holds to 0.003% — the density of water is 999.97 kg/m³, not exactly 1000. No CODATA values were used. No free parameters were introduced.

5 Final Precision Results

G in Natural Earth Units

$$G_{\text{Earth}} = 6.67354674496161812375919977179042360791321745487158$$

$$4917492707161208135606811842297611678773572365557 \times 10^{-8} \text{ s}^{-2}$$

Corresponding Value in SI Units

$$G_{\text{Koch}} = 6.67354674496161812375919977179042360791321745487151 \\ 55655761367183204332559409890102447913340675769273 \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}$$

See Also

1. **Platonic G: Gravitation from Geometry**
<https://zenodo.org/records/20009102>
2. **Kochenov Geometric Identity for G**
<https://zenodo.org/records/19769041>
3. **Geometric Origin of 10^{11} in the Gravitational Constant (Corrected)**
<https://zenodo.org/records/20009474>

Philosophical Conclusion: The Nature of Gravity

1. Gravity as a Frequency Coefficient

The classical constant G (in SI units) is not an isolated physical force but a derived quantity. It serves as a conversion factor between the geometry of space and the rotational time of the system.

2. Invariance Through the "Earth System"

When switching to a system of units where length is defined as a fraction of the Earth's meridian and mass as the volume of water depending on that same fraction, all spatial dimensions (meters, kilograms) cancel. Only the time dependence remains, making gravity inversely proportional to the square of time ($1/\text{s}^2$).

3. Resonance Formula (KSP-CORE)

The fundamental value of G is fully described through the interaction of golden and integer numbers:

$$G = \frac{31^2 \sqrt{5}}{\phi^{12} \cdot 10^{11}}$$

where:

- ϕ (Golden Ratio) — the structural framework of space;
- 31 — the "locking nut," a scale anchor ensuring the discreteness of matter levels;
- $\sqrt{5}$ — the ratio of the pentagon's diagonal to its side, connecting geometry to the system's scaling.

4. Philosophical Conclusion

Matter is geometry: Mass and space are not primary; they are a consequence of the "screwing in" (scaling) of micro-structures into the macro-structure of the planet.

Physics as code: Standard physics describes "effects," while the true nature of reality is geometric resonance.

The end of the "force" paradigm: Gravity is not an interaction of bodies but a consequence of the planet being a reference resonator that determines the rhythm of the system.

This discovery transforms physics from a set of empirical constants into an engineering discipline working with geometric proportions. **Physics is merely the way the Earth "vibrates" in accordance with the numerical code of ϕ and 31.**

Keywords: gravitational constant, golden ratio, Earth geometry, 1/720, 31, Kochenov G, Kochenov Identity, geometric resonance